Fantasy World Cup 2026

Pricing Retro: Why Players Underpriced vs. Teams

Ronjan Sikdar · July 2026 · Data: live Firestore final state + docs/archive/2026/matches.json, ~/code/fakeronjan/games/fantasy-world-cup

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Across the full 264-player / 48-team pool, teams landed almost exactly on the game's built-in **TARGET_ROI = 2.0** pts/$ (aggregate 2.02), while players fell short (aggregate 1.72). But the aggregate hides the real story: player ROI was wildly bimodal – a handful of true stars paid off at 7–8.5x while a large floor-priced bloc never moved. Three things drove it, in order of weight: (1) teams and players were never priced by the same method to begin with, (2) the mid-tournament reprice fix that corrected it was validated once and then frozen for the rest of the run, and (3) the tournament itself ran favorite-heavy and goal-heavy in a way neither pricing model fully absorbed.

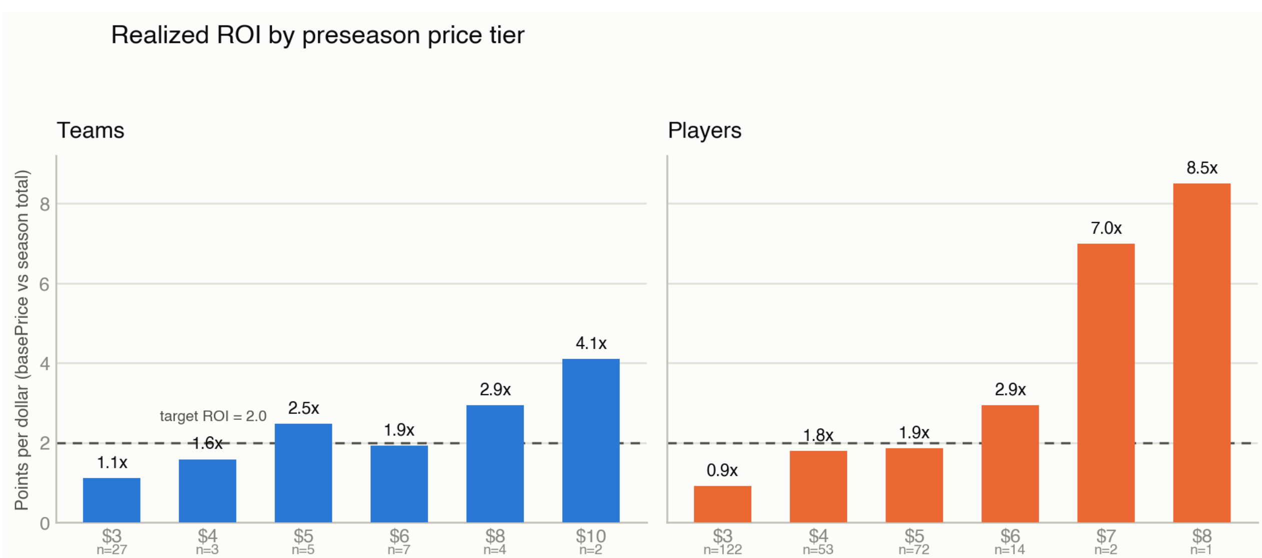
2.02x
TEAM AGGREGATE ROI (target 2.0)

1.72x
PLAYER AGGREGATE ROI (target 2.0)

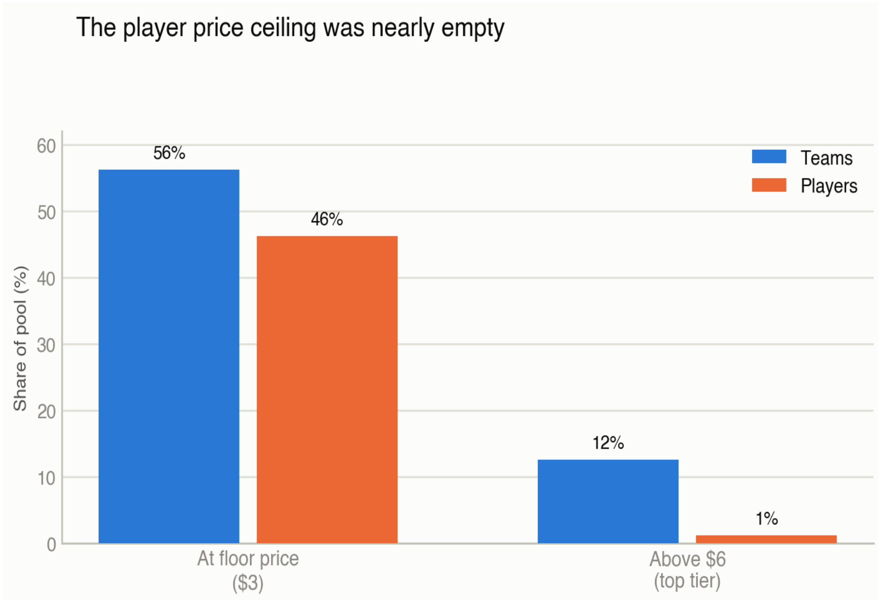
3 (1.1%)
PLAYERS PRICED ABOVE $6 OF 264

1. Realized ROI by price tier

Splitting basePrice into tiers and comparing to season-end totalPoints shows the asymmetry directly. Team ROI rises smoothly from $3 to $10. Player ROI is flat-to-poor through $3–$5 (94% of the pool), then explodes at the very top – the two $7 players and the one $8 player (Mbappé) outperformed even the best-priced team.



basePrice = preseason price. ROI = season-total points ÷ basePrice. n = pool size at that price.



Teams spread across six price tiers with 12% of the pool priced above $6. Players used the same six nominal tiers, but only 1.1% of the player pool (3 of 264) ever cleared $6 – the effective ceiling for a global superstar was barely above the "strong starter" tier.

2. Root causes

2.1 Teams and players were never priced the same way

Team basePrice came from real market signal: Kalshi win-probability odds mapped to a $3–$15 band (commit `3468381`, 2026-05-25) – e.g. France 18.3% implied → $15. Player basePrice came from hand-curated star tiers (`TIER_PRICE = {1:10, 2:7, 3:5, 4:3, 5:2}` in `build_seed_players.py`) plus a rank/position formula, with **no expected-points or ROI anchor at basePrice-setting time at all**. One side of the market started calibrated to a real forecast; the other started as a popularity ranking.

2.2 The reprice fix was validated once, then frozen

The in-tournament reprice engine (`scripts/reprice.py`) targets the same `TARGET_ROI = 2.0` for both kinds, but computes each asset's forward value differently: team value comes from a full Monte Carlo of match outcomes; player value comes from a separate empirical-Bayes goals/game rate model with its own frozen constants. Timeline:

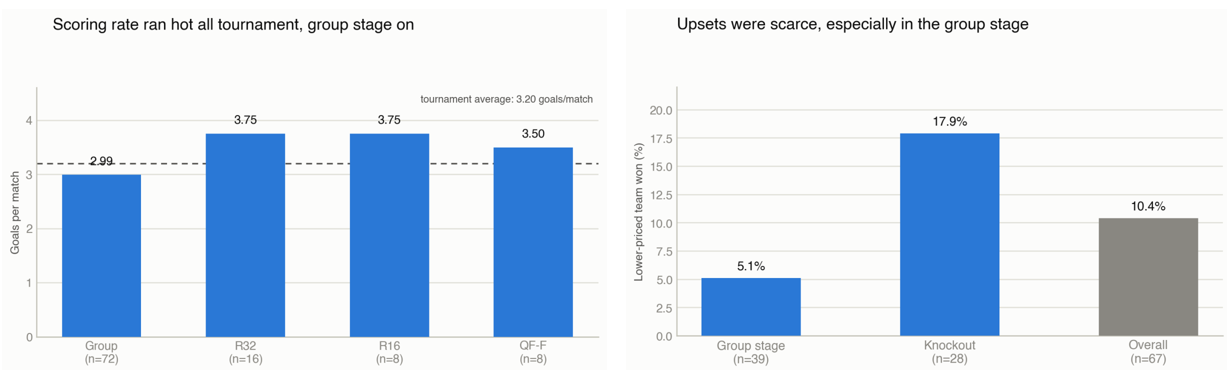
Jun 24	`PR #2 (ca64bd1)`	Form-aware reprice ships (goal-share model). Fixes the "Messi priced at $2" bug.
Jun 26	`be363ec`	The shared match engine (simulate_2026.py) is retuned against 60 real group matches: favorites were winning 69% vs. the model's predicted 45%; goals ran 2.95 vs. predicted 2.6. Feeds team pricing directly.
Jun 28	`a9a7a0e`	Player reprice switches to a per-player goals/game rate model. Commit message: "pts/$ ~1.8 flat across every tier, teams vs players equal value." Declared fixed.

Jun 28 - Jul 19	`(none)`	Zero commits to reprice.py for the rest of the tournament - R16, QF, SF, Final all ran on the Jun 28 constants (FORM_M=3.0, GOAL_PRIOR_SCALE=0.020, GROUP_GAMES=3 hardcoded regardless of how many rounds had actually been played).

The June fixes were real improvements – they solved the original Messi mispricing. But the “fixed” declaration on Jun 28 was validated once, at the R32 boundary, against group-stage-only data, and the underlying match engine's own admission three weeks earlier (favorites winning 24 points more than modeled) was never re-applied to the player-side constants.

2.3 The tournament itself ran favorite-heavy and goal-heavy

Independent of any pricing bug, the real bracket gave favorites – and their star players – more matches and more goals than a typical prior would assume, amplified by the 48-team expansion's weaker new entrants:



Group-stage upset rate (lower-priced team wins a decisive match) was 5.1%; nine matches finished with a 4+ goal margin, almost all favorites routing new 48-team entrants (Qatar 0-6, Curaçao 1-7, Tunisia 0-4/1-5, New Zealand 1-5, Saudi Arabia 0-4, Iraq 0-5, Uzbekistan 0-5). Goals/game stayed elevated into the knockouts (R32/R16 at 3.75) rather than tightening up as rounds got harder.

3. Answering the three hypotheses

Was it the reprice calibration we fixed?

Partly, and only for a while. The Jun 24/28 fixes were real and did solve the worst mispricing (Messi $2). But the fix was declared done after one validation pass and never revisited – it ran unchanged for the last three weeks of the tournament (R16 through the Final) and was never retuned again, while the team-side match engine got its own separate retune on Jun 26 that the player rate model never inherited.

Were there fewer upsets than expected, keeping stars in longer?

Yes, clearly. Group-stage upset rate was 5.1%, well below what the pricing model's own mid-tournament audit assumed (its Jun 26 retune found favorites winning 69% of matches vs. a modeled 45%). Favorites advanced more reliably than priced for, which means their star players simply got more matches, and therefore more points, than any preseason forecast built in.

Were top players scoring more as the tournament expanded to 48 teams?

Yes. Tournament-wide scoring ran at 3.20 goals/match, group stage alone at 2.99, both elevated versus the game's original calibration (2.6 base goals/match). Nine blowouts of 4+ goals, concentrated against the 16 newly-added 48-team entrants, gave elite scorers extra padding right in the group-stage window the reprice model weighted most heavily.

Bottom line

No single cause explains it. The base price curation gave players a structurally thinner ceiling than teams from day one (odds-based vs. hand-tiered, no ROI anchor). The one mid-tournament reprice fix genuinely helped, but was validated once and then left untouched for three more weeks while real results kept diverging from its priors. And the tournament itself – fewer upsets, more goals, both stretched by the 48-team field's weaker new entrants – kept outrunning whatever assumption was frozen in place. The fix for next time isn't one knob: it's giving player basePrice the same odds-style rigor teams got, and re-validating the reprice engine at every round boundary instead of once.

Sources: live Firestore dump (teams/players basePrice + totalPoints, 2026-07-22), docs/archive/2026/matches.json (104 matches, final scores), scripts/reprice.py, scripts/build_seed_players.py, scripts/simulate_2026.py, and repo commit history in ~/code/fakeronjan/games/fantasy-world-cup.